RAJ BHARAT RASAL

Distributed Systems & Backend Architect

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PROFESSIONAL SUMMARY

Second-year B.Tech CSE student specializing in distributed systems architecture, tensor sharding, and zero-trust backend infrastructure. Founder of SecureVerse with practical experience building distributed compute clusters and resilient network protocols.

TECHNICAL & ARCHITECTURAL SKILLS

Distributed Systems & Architecture: Distributed Systems Architecture, Tensor Sharding & Compute Networks, Decoupled Orchestration, Zero-Trust Communication Protocols, Proof-of-Compute Verification, System Design & Scalability, Futuristic Systems Design

Backend & Cloud: FastAPI, RESTful API Design, Asynchronous Programming (asyncio), APScheduler & Task Queues, Tailscale VPN & Mesh Networking, Redis Streams & Message Passing

Security & Systems: Vulnerability Research & Bug Bounty, Digital Forensics & Reverse Engineering, Zero-Trust Data Flow Verification, Sandboxing & Isolation (Firecracker)

FEATURED SYSTEMS & PROJECTS

SATURDAY / SUNDAY — Architect & Lead Developer

Jan 2026 - Present

Tech Stack: Python, C++, Tensor Sharding, Zero-Trust Protocols, LAN, Tailscale VPN, Socket Programming

- Built a distributed compute network to run large local LLMs, solving hardware limitations of processing complex AI models on single devices.
- Implemented automatic hardware discovery and dynamic tensor sharding to split a single model's processing across heterogeneous Mac and Windows nodes via LAN and Tailscale VPN mesh.
- Designed a zero-trust communication protocol between nodes, utilizing proof-of-compute verification to detect and reject nodes returning corrupted or falsified matrix calculation results.
- **Key Architecture:** Decoupled coordinator-worker model with heartbeats; distributed matrix multiplication with Tailscale mesh tunnels; sub-millisecond node health checks; Byzantine fault tolerance through spot verification.

F.R.I.D.A.Y — Multi-Agent Communication Backend — Lead Backend Architect & Developer

Jun 2026 - Present

Tech Stack: Python, FastAPI, Baileys API (WhatsApp), APScheduler, asyncio, LLM Intent Routing

- Originated at SummerHacks 2026 in a team of 4; currently driving continuous backend evolution and multi-agent infrastructure.
- Designed a strictly decoupled 3-agent architecture (Herald, Nexus, Dispatch) enforcing complete separation of concerns so agents never call each other directly, eliminating circular deadlocks.
- Implemented LLM intent classification with confidence-based routing logic to catch ambiguous queries and trigger clarification flows before dispatching execution tasks.
- Engineered resilient background job scheduling and automated task pipelines with robust fault recovery.
- **Key Architecture:** 3-tier mediator topology; zero direct agent coupling; confidence threshold routing (<0.85 triggers conversational disambiguation); asynchronous message queue via FastAPI.

SecureVerse — Decentralized Zero-Trust Architecture — Founder & Systems Architect

Oct 2025 - Dec 2025

Tech Stack: Python, Cryptography, Zero-Trust Architecture, Distributed Ledger, REST APIs

- Architected a decentralized identity and zero-trust verification platform engineered to prevent credential theft and unauthorized access in enterprise networks.
- Formulated the end-to-end cryptographic trust chain, policy evaluation engine, and auditable logging system.
- Awarded 3rd Place at the Deal Dome Competition for superior architectural soundness and threat mitigation design.
- **Key Architecture:** Mutual TLS, cryptographic assertion validation, state machine based authorization engine, zero persistent plain text secrets.

EXPERIENCE & LEADERSHIP

SecureVerse — Founder & Lead Architect

Oct 2025 - Present

- Founded SecureVerse to engineer decentralized zero-trust identity verification architectures, AI safety tooling, and privacy-preserving computing systems.
- Formulated end-to-end cryptographic trust chains, automated audit mechanisms, and threat mitigation models recognized in national engineering challenges.
- Directed technology roadmap, leading proof-of-concept development and cross-functional prototype verification.

Independent Tutoring Practice — Mathematics & Physics Tutor (Grades 10–12) *Jun 2024 - Present*

- Taught 50+ high school students, translating complex scientific, mathematical, and algorithmic concepts into clear, structured lessons.
- Developed diagnostic assessment frameworks to identify conceptual bottlenecks and tailored individual curricula to improve performance by an average of 22%.
- Honed mastery of first-principles thinking, communication clarity, and pedagogical system design.

SYSTEM DESIGN SPECIFICATIONS & WHITEPAPERS

HAMIA Architecture Specification: Hierarchical Adaptive Multi-Model Intelligence Architecture (Architecture Whitepaper, Oct 2026)

- A cloud-native fusion system that orchestrates a federation of specialized frontier models (12+) through an 8-stage pipeline. Emphasizes evidence-centric fusion over naive voting, using external ground-truth verifiers (Firecracker sandboxes, SymPy) and a cyclic adversarial critic to prevent multi-model ensembles from amplifying shared bias.

EDUCATION

ITM Skills University — Bachelor of Technology (B.Tech) *Aug 2025 - May 2029*

Field: Computer Science Engineering | Score: Undergraduate (In Progress)

- Specializing in Distributed Systems, Operating Systems, Computer Networks, and AI Systems Design. Active contributor to student engineering and hackathon teams.

Central Board of Secondary Education (CBSE) — Higher Secondary Certificate (12th Grade) *2023 - 2025*

Field: Science (PCM) | Score: 83.4%

- Strong foundation in advanced calculus, mechanics, electromagnetic physics, and algorithmic logic.

Central Board of Secondary Education (CBSE) — Secondary School Examination (10th Grade) *2021 - 2023*

Field: General Science & Mathematics | Score: 96.0%

- Academic excellence award in Mathematics and Science.

ACHIEVEMENTS & SECURITY RESEARCH

- **3rd Place — Deal Dome Competition:** Awarded 3rd place for designing the architectural specification and threat modeling for SecureVerse, a zero-trust enterprise security framework.
- **Security Research: OPPO Find X3 Pro Vulnerability:** Identified and responsibly reported an Open Redirect vulnerability in the OPPO Find X3 Pro flagship device software ecosystem, acknowledged in HackerOne Hall of Fame.

CERTIFICATIONS & ACCREDITATIONS

- **Advanced Backend Engineering & Distributed Systems Architecture** — Engineering Systems Council (2025)
- **Responsible Disclosure & Security Research Recognition** — HackerOne (2025)